Changes in Adolescent Alcohol Use

Group A

2026-03-07

Data Summary

```
alc <- read.table("alcohol.txt", header=TRUE) # loading the dataset
alc <- alc %>%
  mutate(coa = factor(coa, levels=c("0", "1")), # convert parental alcoholism to a factor
                      male=factor(male, levels=c("0", "1"))) # convert gender to a factor

head(alc)
```

```
##   id age coa male   alcuse      peer
## 1  1  14   1    0 1.732051 1.2649111
## 2  1  15   1    0 2.000000 1.2649111
## 3  1  16   1    0 2.000000 1.2649111
## 4  2  14   1    1 0.000000 0.8944272
## 5  2  15   1    1 0.000000 0.8944272
## 6  2  16   1    1 1.000000 0.8944272
```

```
alc_base <- alc %>% # extract baseline observations (age 14)
  group_by(id) %>%
  slice(1) %>%
  ungroup()

summary_table <- bind_rows(

  tibble(
    Characteristic = "Alcohol use",
    Value = ""
  ),

  # summarizing alcohol use for each age
  bind_rows(lapply(sort(unique(alc$age)), function(a) {
    dat <- alc %>% filter(age == a)
    tibble(
      Characteristic = c(
        paste0("\\hspace{1em}Age ", a),
        "\\hspace{2em}Mean (SD)",
        "\\hspace{2em}Median",
        "\\hspace{2em}Range"
      ),
      Value = c(
```

```r
      "",
      sprintf("%.2f (%.2f)", mean(dat$alcuse, na.rm = TRUE), sd(dat$alcuse, na.rm = TRUE)),
      sprintf("%.2f", median(dat$alcuse, na.rm = TRUE)),
      sprintf("%.2f--%.2f", min(dat$alcuse, na.rm = TRUE), max(dat$alcuse, na.rm = TRUE))
    )
  )
}),

tibble( # creating a tibble to summarize peer alcohol use
  Characteristic = c(
    "Peer alcohol use",
    "\\hspace{1em}Mean (SD)",
    "\\hspace{1em}Median",
    "\\hspace{1em}Range"
  ),
  Value = c(
    "",
    sprintf("%.2f (%.2f)", mean(alc_base$peer, na.rm = TRUE), sd(alc_base$peer, na.rm = TRUE)),
    sprintf("%.2f", median(alc_base$peer, na.rm = TRUE)),
    sprintf("%.2f--%.2f", min(alc_base$peer, na.rm = TRUE), max(alc_base$peer, na.rm = TRUE))
  )
),

# summarize the proportion of each gender
alc_base %>%
  mutate(
    male_label = case_when(
      male == "0" ~ "Female",
      male == "1" ~ "Male",
      TRUE ~ as.character(male)
    )
  ) %>%
  count(male_label) %>%
  mutate(
    Characteristic = paste0("\\hspace{1em}", male_label),
    Value = paste0(n, " (", round(100 * n / sum(n), 1), "\\%)")
  ) %>%
  select(Characteristic, Value) %>%
  bind_rows(tibble(Characteristic = "Gender", Value = ""), .),

# summarizing the proportion of COA = 0 vs 1
alc_base %>%
  mutate(
    coa_label = case_when(
      coa == "0" ~ "No",
      coa == "1" ~ "Yes",
      TRUE ~ as.character(coa)
    )
  ) %>%
  count(coa_label) %>%
  mutate(
    Characteristic = paste0("\\hspace{1em}", coa_label),
    Value = paste0(n, " (", round(100 * n / sum(n), 1), "\\%)")
```

```r
    ) %>%
    select(Characteristic, Value) %>%
    bind_rows(tibble(Characteristic = "Parental alcoholism", Value = ""), .)
)

kable(
  summary_table,
  format = "latex",
  booktabs = TRUE,
  caption = "Summary of study variables",
  col.names = c("Variables", "Value"),
  align = "ll",
  linesep = "",
  escape = FALSE
) %>%
  kable_styling(
    latex_options = "hold_position",
    font_size = 10
  ) %>% landscape()
```

Table 1: Summary of study variables

Variables	Value
Alcohol use	
Age 14	
Mean (SD)	0.63 (0.94)
Median	0.00
Range	0.00–3.32
Age 15	
Mean (SD)	0.96 (1.05)
Median	1.00
Range	0.00–3.61
Age 16	
Mean (SD)	1.17 (1.14)
Median	1.00
Range	0.00–3.46
Peer alcohol use	
Mean (SD)	1.02 (0.73)
Median	0.89
Range	0.00–2.53
Gender	
Female	40 (48.8%)
Male	42 (51.2%)
Parental alcoholism	
No	45 (54.9%)
Yes	37 (45.1%)

Missing data

```
missing_table <- alc %>% # summarize missing data by age
  group_by(age) %>%
  summarise(
    "Number of observations" = n(),
    "Alcohol use" = sum(is.na(alcuse)),
    "Peer alcohol use" = sum(is.na(peer)),
    "Parental alcoholism" = sum(is.na(coa)),
    "Gender" = sum(is.na(male))
  )

paper_kable(missing_table, "Missing data summary by age")
```

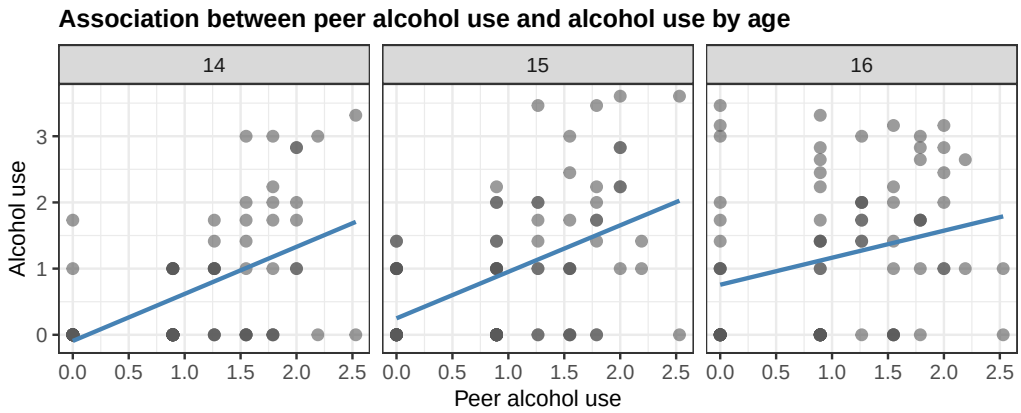
Table 2: Missing data summary by age

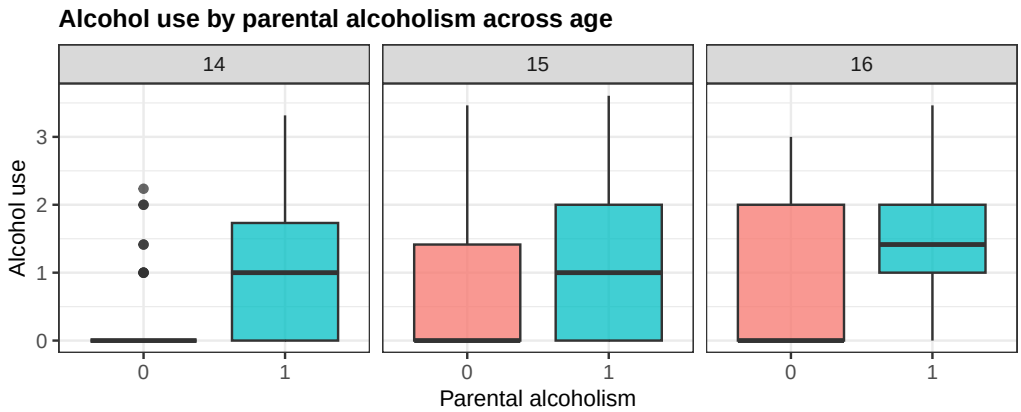
age	Number of observations	Alcohol use	Peer alcohol use	Parental alcoholism	Gender
14	82	0	0	0	0
15	82	0	0	0	0
16	82	0	0	0	0

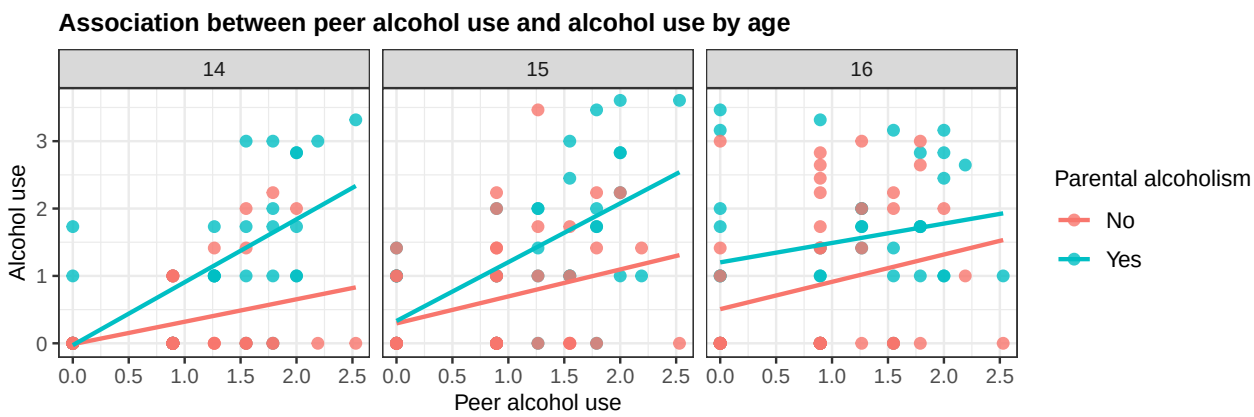
Exploratory plots

```
# creating plots to visualize relations
p_box_age <- ggplot(alc, aes(x = coa, y = alcuse, fill = coa)) + # boxplot
  geom_boxplot(alpha = 0.75, show.legend = FALSE) +
  facet_wrap(~ age) +
  scale_fill_manual(values = coa_cols) +
  labs(
    title = "Alcohol use by parental alcoholism across age",
    x = "Parental alcoholism",
    y = "Alcohol use"
  ) +
  base_plot_theme

p_peer_age <- ggplot(alc, aes(x = peer, y = alcuse)) + # scatterplot
  geom_point(alpha = 0.6, size = 2, color = "grey35") +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.9, color = "steelblue") +
  facet_wrap(~ age) +
  labs(
    title = "Association between peer alcohol use and alcohol use by age",
    x = "Peer alcohol use",
    y = "Alcohol use"
  ) +
  base_plot_theme

p_peer_age_coa <- ggplot(alc, aes(x = peer, y = alcuse, color = coa)) + # scatterplot by coa
  geom_point(alpha = 0.8, size = 2) +
```

```r
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.9) +
  facet_wrap(~ age) +
  labs(
    title = "Association between peer alcohol use and alcohol use by age",
    x = "Peer alcohol use",
    y = "Alcohol use",
    color = "Parental alcoholism"
  ) +
  scale_color_manual(
    values = coa_cols,
    labels = c("No", "Yes")
  ) +
  base_plot_theme

# grouping the tables onto the same page
eda_plots <- (
  p_peer_age +
  p_box_age +
  p_peer_age_coa
) +
  plot_layout(ncol = 1)

eda_plots
```



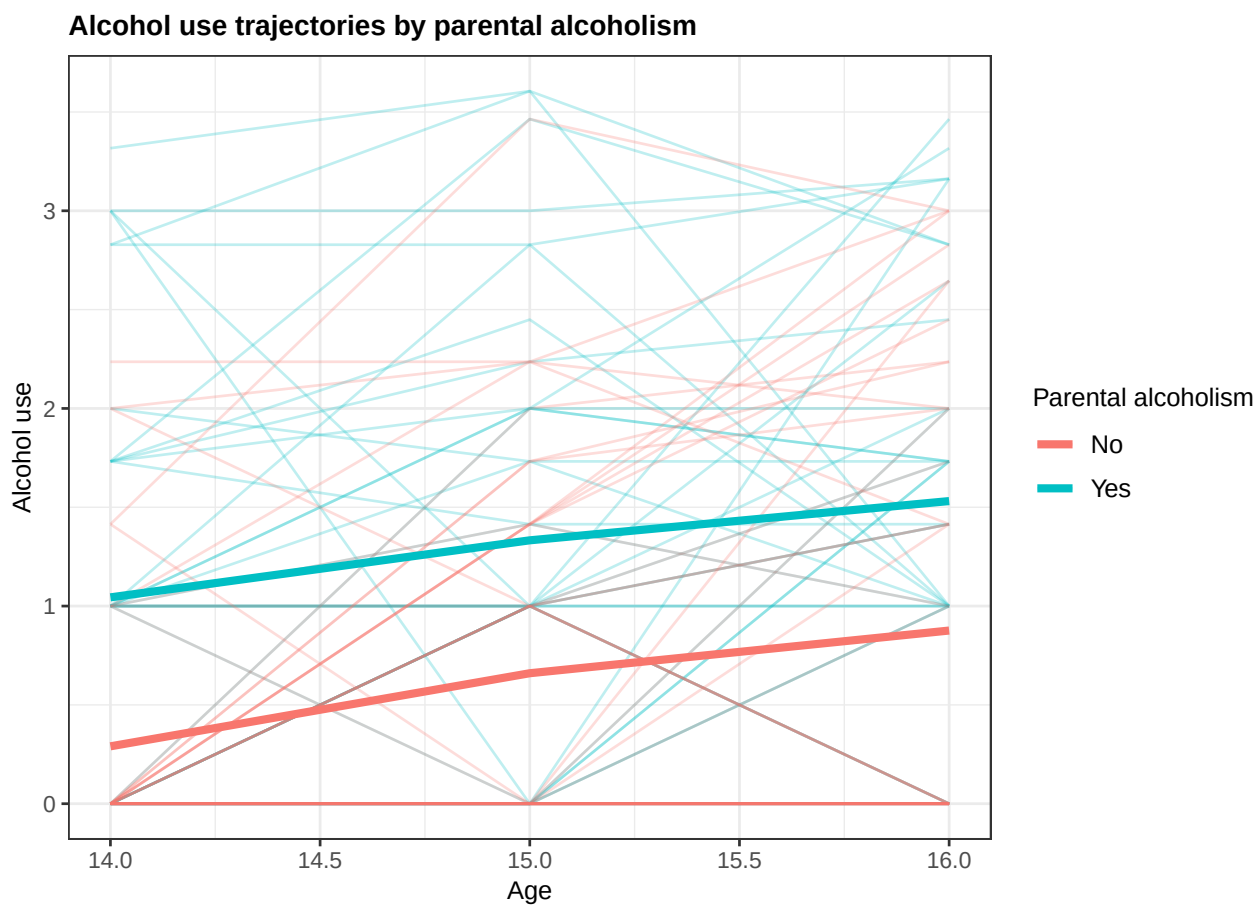




Individual trajectories with mean trend

```
# plotting the trajectory plot
ggplot(alc, aes(x = age, y = alcuse, group = id, color = coa)) +
  geom_line(alpha = 0.25) +
  stat_summary(
    aes(group = coa, color = coa),
    fun = mean,
    geom = "line",
    linewidth = 1.5
  ) +
  scale_color_manual(values = coa_cols, labels = c("No", "Yes"),
```

```
                      name = "Parental alcoholism") +
  labs(
    title = "Alcohol use trajectories by parental alcoholism",
    x = "Age",
    y = "Alcohol use"
  ) +
  base_plot_theme
```



Model comparison

```
# we choose the use linear mixed effect model since we are interested in individual trajectories
# we use the full model to choose the covariance structure
m1 <- lme( # random intercept + ind errors model
  alcuse ~ age * coa * peer + male,
  random = ~ 1 | id,
  data = alc,
  method = "REML"
)

m2 <- lme( # random intercept and slopes + ind errors model
  alcuse ~ age * coa * peer + male,
  random = ~ age | id,
```

```r
  data = alc,
  method = "REML"
)

m3 <- lme( # random intercept + AR1 errors model
  alcuse ~ age * coa * peer + male,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "REML"
)


m4 <- lme( # random intercept + heteroske ind errors (age) model
  alcuse ~ age * coa * peer + male,
  random = ~ 1 | id,
  weights = varIdent(form = ~ 1 | age),
  data = alc,
  method = "REML"
)

m5 <- lme( # random intercept + heteroske ind errors (gender) model
  alcuse ~ age * coa * peer + male,
  random = ~ 1 | id,
  weights = varIdent(form = ~ 1 | male),
  data = alc,
  method = "REML"
)

cov_table <- tibble(
  `Covariance model` = c(
    "Random intercepts + independent errors",
    "Random intercepts and slopes + independent errors",
    "Random intercepts + AR(1) errors",
    "Random intercepts + heteroskedastic independent errors by age",
    "Random intercepts + heteroskedastic independent errors by gender"
  ),
  AIC = c(AIC(m1), AIC(m2), AIC(m3), AIC(m4), AIC(m5)),
  BIC = c(BIC(m1), BIC(m2), BIC(m3), BIC(m4), BIC(m5))
) %>%
  mutate(
    AIC = round(AIC, 3),
    BIC = round(BIC, 3)
  )

paper_kable(cov_table, "Comparison of covariance structures")
```

Random intercept and slope model has the lowest AIC and BIC.

Table 3: Comparison of covariance structures

Covariance model	AIC	BIC
Random intercepts + independent errors	640.750	678.899
Random intercepts and slopes + independent errors	626.169	671.254
Random intercepts + AR(1) errors	634.744	676.360
Random intercepts + heteroskedastic independent errors by age	634.656	679.741
Random intercepts + heteroskedastic independent errors by gender	641.983	683.600

Random intercept and slope model

Fixed effect model selection

```
m_mean1 <- lme( # main effect model
  alcuse ~ age + coa + peer,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean2 <- lme( # main effect model with gender
  alcuse ~ age + coa + peer + male,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean3 <- lme( # two way interaction model
  alcuse ~ age + coa + peer + age:coa + age:peer + coa:peer,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean4 <- lme( # two way interaction model with gender
  alcuse ~ age + coa + peer + male + age:coa + age:peer + coa:peer + male:age + male:peer + male:coa,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean5 <- lme( # three way interaction model
  alcuse ~ age * coa * peer,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

m_mean6 <- lme( # three way interaction model with gender
  alcuse ~ age * coa * peer * male,
```

```r
  random = ~ age | id,
  data = alc,
  method = "ML"
)
```

```r
# without gender model comparison using anova()
mean_model_compare <- anova(m_mean1, m_mean3, m_mean5) %>%
  as.data.frame() %>%
  mutate(
    Model = c(
      "1. Main effects",
      "2. Main effects + two-way interactions",
      "3. Main effects + two-way + three-way interactions"
    )
  ) %>%
  select(Model, df, logLik, `L.Ratio`, `p-value`, `Test`) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

rownames(mean_model_compare) <- NULL

# writing the model output into a table
paper_kable(
  mean_model_compare,
  "Likelihood ratio tests for fixed-effects structure selection without gender"
)
```

Table 4: Likelihood ratio tests for fixed-effects structure selection without gender

Model	df	logLik	L.Ratio	p-value	Test
1. Main effects	8	-295.925	NA	NA	
2. Main effects + two-way interactions	11	-291.296	9.259	0.026	1 vs 2
3. Main effects + two-way + three-way interactions	12	-289.038	4.516	0.034	2 vs 3

For without gender, three way interaction is the best.

```r
# with gender model comparison using anova()
mean_model_compare <- anova(m_mean2, m_mean4, m_mean6) %>%
  as.data.frame() %>%
  mutate(
    Model = c(
      "1. Main effects",
      "2. Main effects + two-way interactions",
      "3. Main effects + two-way + three-way interactions"
    )
  ) %>%
  select(Model, df, logLik, `L.Ratio`, `p-value`, `Test`) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

rownames(mean_model_compare) <- NULL

# writing the model output into a table
```

```r
paper_kable(
  mean_model_compare,
  "Likelihood ratio tests for fixed-effects structure selection with gender"
)
```

Table 5: Likelihood ratio tests for fixed-effects structure selection with gender

Model	df	logLik	L.Ratio	p-value	Test
1. Main effects	9	-295.347	NA	NA	
2. Main effects + two-way interactions	15	-287.338	16.018	0.014	1 vs 2
3. Main effects + two-way + three-way interactions	20	-283.782	7.111	0.213	2 vs 3

For with gender, two way interaction is the best.

```r
# AIC and BIC to compare the two models
mean_table <- tibble(
  `Model` = c(
    "Two-way interactions + gender",
    "Three-way interactions"
  ),
  AIC = c(
    AIC(m_mean4), AIC(m_mean5)
  ),
  BIC = c(
    BIC(m_mean4), BIC(m_mean5)
  )
) %>%
  mutate(
    AIC = round(AIC, 3),
    BIC = round(BIC, 3)
  )

paper_kable(mean_table, "Model Comparison")
```

Table 6: Model Comparison

Model	AIC	BIC
Two-way interactions + gender	604.675	657.255
Three-way interactions	602.075	644.139

Three way interaction without gender is the best.

```r
# likelihood ratio tests
lrt_no_gender <- anova(m_mean1, m_mean3, m_mean5) %>%
  as.data.frame()

lrt_with_gender <- anova(m_mean2, m_mean4, m_mean6) %>%
  as.data.frame()
```

```r
# one combined table
mean_model_compare_all <- bind_rows(
  tibble(
    Model = c(
      "1. Main effects (no gender)",
      "2. Two-way interactions (no gender)",
      "3. Three-way interactions (no gender)"
    ),
    Test = c("", "1 vs 2", "2 vs 3"),
    df = lrt_no_gender$df,
    logLik = lrt_no_gender$logLik,
    `L.Ratio` = lrt_no_gender$`L.Ratio`,
    `p-value` = lrt_no_gender$`p-value`,
    AIC = c(AIC(m_mean1), AIC(m_mean3), AIC(m_mean5)),
    BIC = c(BIC(m_mean1), BIC(m_mean3), BIC(m_mean5))
  ),
  tibble(
    Model = c(
      "4. Main effects (with gender)",
      "5. Two-way interactions (with gender)",
      "6. Three-way interactions (with gender)"
    ),
    Test = c("", "4 vs 5", "5 vs 6"),
    df = lrt_with_gender$df,
    logLik = lrt_with_gender$logLik,
    `L.Ratio` = lrt_with_gender$`L.Ratio`,
    `p-value` = lrt_with_gender$`p-value`,
    AIC = c(AIC(m_mean2), AIC(m_mean4), AIC(m_mean6)),
    BIC = c(BIC(m_mean2), BIC(m_mean4), BIC(m_mean6))
  )
) %>%
  mutate(
    across(c(logLik, `L.Ratio`, `p-value`, AIC, BIC), ~ round(.x, 3))
  )

paper_kable(
  mean_model_compare_all,
  "Likelihood ratio tests and information criteria for fixed-effects structure selection."
)
```

Table 7: Likelihood ratio tests and information criteria for fixed-effects structure selection.

Model	Test	df	logLik	L.Ratio	p-value	AIC	BIC
1. Main effects (no gender)		8	-295.925	NA	NA	607.850	635.893
2. Two-way interactions (no gender)	1 vs 2	11	-291.296	9.259	0.026	604.592	643.150
3. Three-way interactions (no gender)	2 vs 3	12	-289.038	4.516	0.034	602.075	644.139
4. Main effects (with gender)		9	-295.347	NA	NA	608.694	640.242
5. Two-way interactions (with gender)	4 vs 5	15	-287.338	16.018	0.014	604.675	657.255
6. Three-way interactions (with gender)	5 vs 6	20	-283.782	7.111	0.213	607.564	677.671

```r
final_model_int_slope <- lme( # three way interaction model
  alcuse ~ age * coa * peer,
  random = ~ age | id,
  data = alc,
  method = "ML"
)

# Try a knot at age 15 (change in slope on trajectory plots)
final_model_int_slope_knot <- lme(
  alcuse ~ age * coa * peer + age15*coa*peer,
  random = ~ age | id,
  data = alc %>% mutate(age15 = I(age >=15)),
  method = "ML"
)

anova(final_model_int_slope_knot, final_model_int_slope)
```

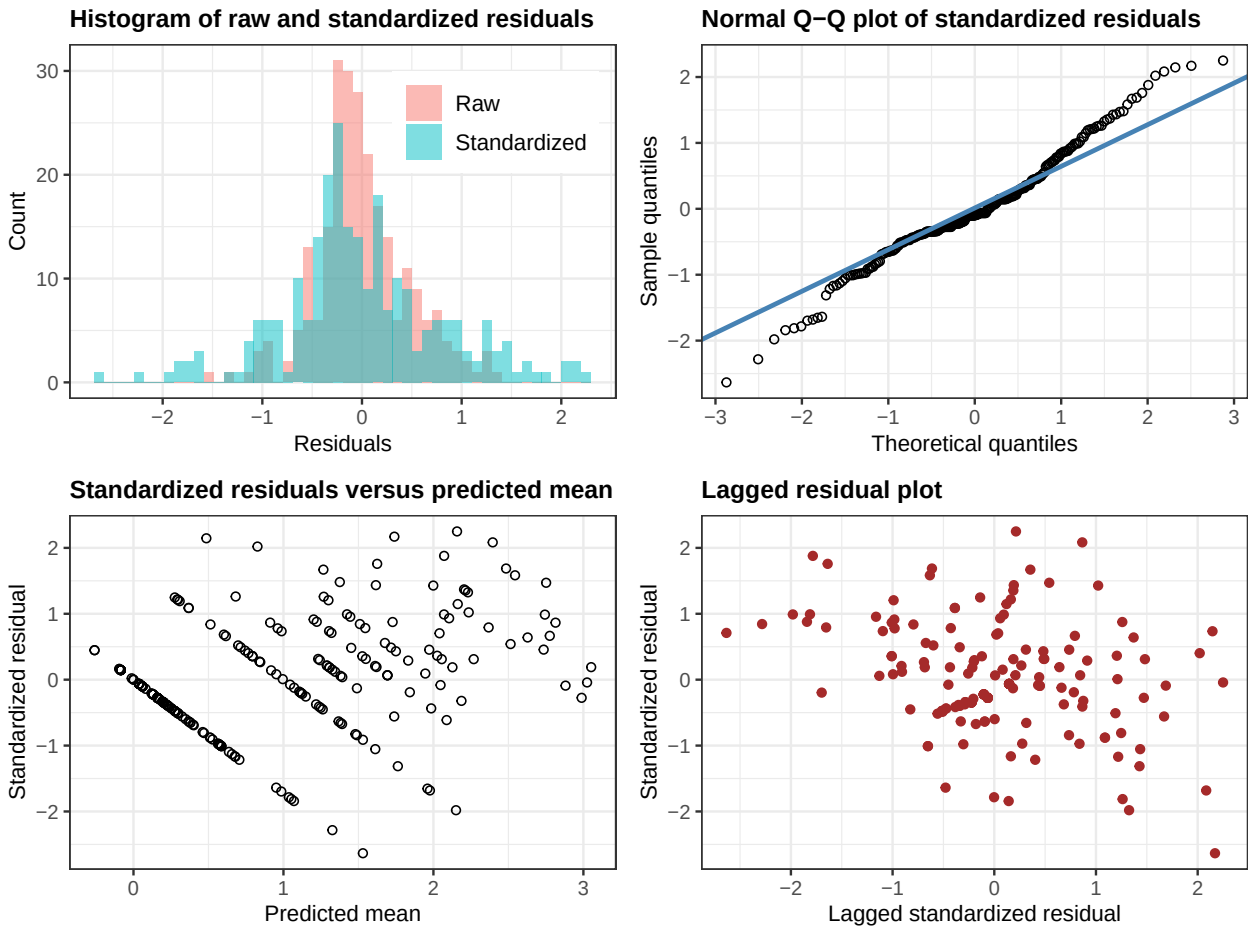
```
##                             Model df      AIC      BIC    logLik   Test  L.Ratio
## final_model_int_slope_knot      1 16 606.3386 662.4239 -287.1693
## final_model_int_slope           2 12 602.0753 644.1393 -289.0377 1 vs 2 3.736765
##                             p-value
## final_model_int_slope_knot
## final_model_int_slope        0.4428
```

```r
# Non-significant - continue without a knot
```

Model diagnostic

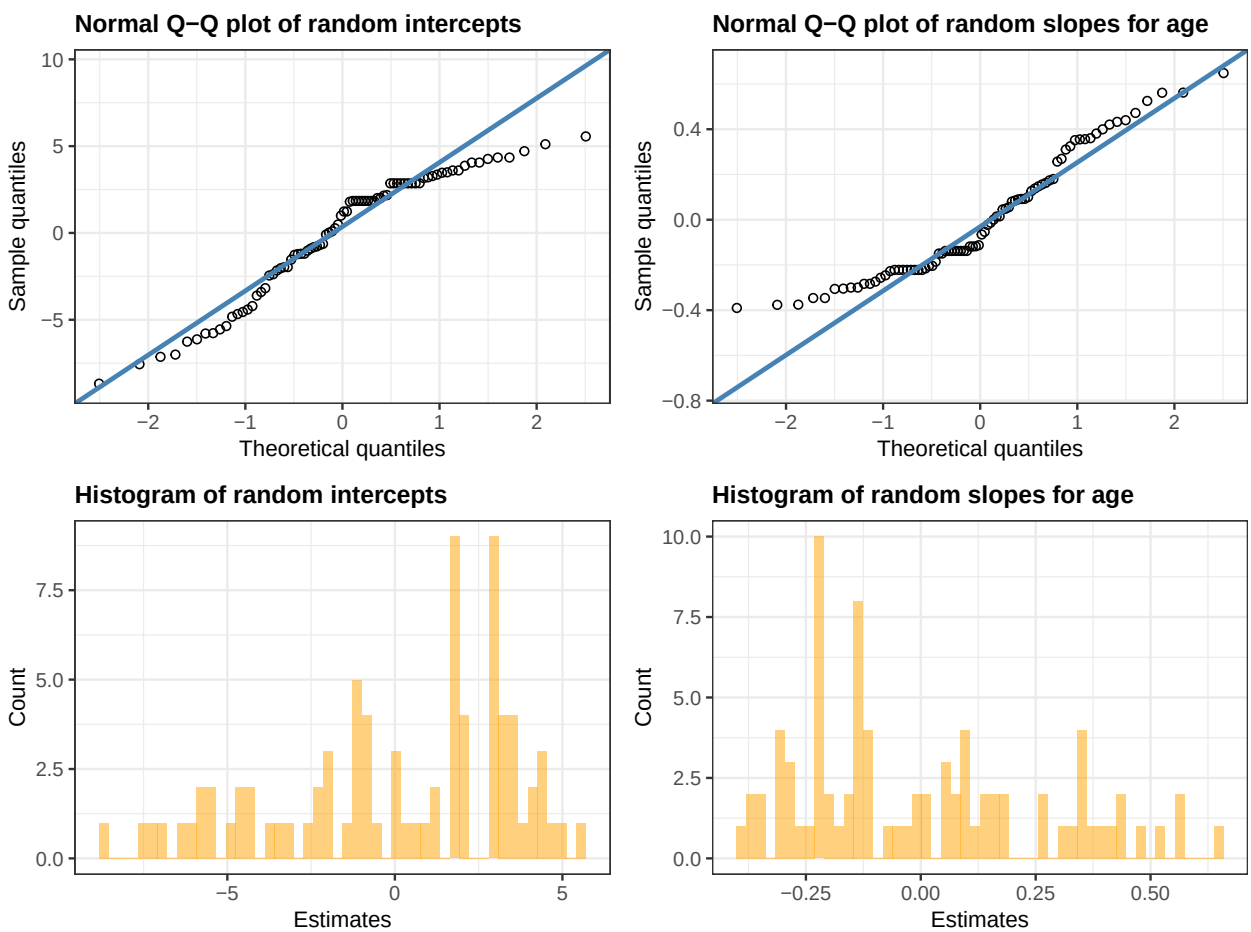
```r
rhat <- resid(final_model_int_slope, type = "normalized")

condmean <- predict(final_model_int_slope)
y <- alc$alcuse
stg1r <- y - condmean

rtype <- c(rep("Raw", length(stg1r)), rep("Standardized", length(stg1r)))
resid <- c(stg1r, rhat)

stg1resids <- data.frame(
  rtype = rtype,
  resid = as.numeric(resid)
)

p1 <- ggplot(stg1resids, aes(x = resid, fill = rtype)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50) +
  labs(
    title = "Histogram of raw and standardized residuals",
    y = "Count",
    x = "Residuals",
    fill = NULL
  ) +
  theme(
```

```r
    legend.position = c(0.78, 0.78)
  ) +
  base_plot_theme

p2 <- ggplot(data.frame(rhat = rhat), aes(sample = rhat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of standardized residuals",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

yhat <- predict(final_model_int_slope)

p3 <- ggplot(data.frame(yhat = yhat, rhat = rhat), aes(y = rhat, x = yhat)) +
  geom_point(shape = 1) +
  labs(
    title = "Standardized residuals versus predicted mean",
    x = "Predicted mean",
    y = "Standardized residual"
  ) +
  base_plot_theme

p4 <- alc %>%
  mutate(rhat = rhat) %>%
  group_by(id) %>%
  mutate(rhat1 = lag(rhat)) %>%
  ggplot(aes(y = rhat, x = rhat1)) +
  geom_point(color = "brown", na.rm = TRUE) +
  labs(
    title = "Lagged residual plot",
    y = "Standardized residual",
    x = "Lagged standardized residual"
  ) +
  base_plot_theme

(p1 + p2) / (p3 + p4)
```




```
re <- ranef(final_model_int_slope) %>%
  rownames_to_column("id")

b0hat <- re$`(Intercept)`
b1hat <- re$age

p1 <- ggplot(data.frame(b0hat = b0hat), aes(sample = b0hat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of random intercepts",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

p2 <- ggplot(data.frame(b1hat = b1hat), aes(sample = b1hat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of random slopes for age",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
```

```r
  base_plot_theme

p3 <- ggplot(data.frame(b0hat = b0hat), aes(x = b0hat)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50, fill = "orange") +
  labs(
    title = "Histogram of random intercepts",
    x = "Estimates",
    y = "Count"
  ) +
  base_plot_theme

p4 <- ggplot(data.frame(b1hat = b1hat), aes(x = b1hat)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50, fill = "orange") +
  labs(
    title = "Histogram of random slopes for age",
    x = "Estimates",
    y = "Count"
  ) +
  base_plot_theme

(p1 + p2) / (p3 + p4)
```



The residual plots show some violations, so we tried the second best model from the random-effects model selection.

Random intercept only model + AR(1) error

Fixed effect model selection

```
m_mean1 <- lme( # mean effect model
  alcuse ~ age + coa + peer,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "ML"
)

m_mean2 <- lme( # mean effect model with gender
  alcuse ~ age + coa + peer + male,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "ML"
)

m_mean3 <- lme( # two way interaction model
  alcuse ~ age + coa + peer + age:coa + age:peer + coa:peer,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "ML"
)

m_mean4 <- lme( # two way interaction model with gender
  alcuse ~ age + coa + peer + male + age:coa + age:peer + coa:peer + male:age + male:peer + male:coa,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "ML"
)

m_mean5 <- lme( # three way interaction model
  alcuse ~ age * coa * peer,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "ML"
)

m_mean6 <- lme( # three way interaction model with gender
  alcuse ~ age * coa * peer * male,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "ML"
)
```

```r
# without gender model comparison using anova()
mean_model_compare <- anova(m_mean1, m_mean3, m_mean5) %>%
  as.data.frame() %>%
  mutate(
    Model = c(
      "1. Main effects",
      "2. Main effects + two-way interactions",
      "3. Main effects + two-way + three-way interactions"
    )
  ) %>%
  select(Model, df, logLik, `L.Ratio`, `p-value`, `Test`) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

rownames(mean_model_compare) <- NULL

# writing the model output into a table
paper_kable(
  mean_model_compare,
  "Likelihood ratio tests for fixed-effects structure selection without gender"
)
```

Table 8: Likelihood ratio tests for fixed-effects structure selection without gender

Model	df	logLik	L.Ratio	p-value	Test
1. Main effects	7	-300.888	NA	NA	
2. Main effects + two-way interactions	10	-298.333	5.110	0.164	1 vs 2
3. Main effects + two-way + three-way interactions	11	-296.021	4.624	0.032	2 vs 3

```r
# we can see that the two way interaction shows no improvement to the main effect model
# but the three way interaction shows evidence against H0
# anova(m_mean1, m_mean5) shows p-value < 0.05
# provides further evidence that three-way interactions is the best model
```

For without gender, three way interaction is the best.

```r
# with gender model comprison using anova()
mean_model_compare <- anova(m_mean2, m_mean4, m_mean6) %>%
  as.data.frame() %>%
  mutate(
    Model = c(
      "1. Main effects",
      "2. Main effects + two-way interactions",
      "3. Main effects + two-way + three-way interactions"
    )
  ) %>%
  select(Model, df, logLik, `L.Ratio`, `p-value`, `Test`) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

rownames(mean_model_compare) <- NULL
```

```r
# writing the model output into a table
paper_kable(
  mean_model_compare,
  "Likelihood ratio tests for fixed-effects structure selection with gender"
)
```

Table 9: Likelihood ratio tests for fixed-effects structure selection with gender

Model	df	logLik	L.Ratio	p-value	Test
1. Main effects	8	-299.052	NA	NA	
2. Main effects + two-way interactions	14	-292.472	13.160	0.041	1 vs 2
3. Main effects + two-way + three-way interactions	19	-288.797	7.349	0.196	2 vs 3

For with gender, two way interaction is the best.

```r
# AIC and BIC to compare the two models
mean_table <- tibble(
  `Model` = c(
    "Two-way interactions + gender",
    "Three-way interactions"
  ),
  AIC = c(
    AIC(m_mean4), AIC(m_mean5)
  ),
  BIC = c(
    BIC(m_mean4), BIC(m_mean5)
  )
) %>%
  mutate(
    AIC = round(AIC, 3),
    BIC = round(BIC, 3)
  )

paper_kable(mean_table, "Model Comparison")
```

Table 10: Model Comparison

Model	AIC	BIC
Two-way interactions + gender	612.943	662.018
Three-way interactions	614.043	652.602

```r
# likelihood ratio tests
lrt_no_gender <- anova(m_mean1, m_mean3, m_mean5) %>%
  as.data.frame()

lrt_with_gender <- anova(m_mean2, m_mean4, m_mean6) %>%
  as.data.frame()

# one combined table
```

```r
mean_model_compare_all <- bind_rows(
  tibble(
    Model = c(
      "1. Main effects (no gender)",
      "2. Two-way interactions (no gender)",
      "3. Three-way interactions (no gender)"
    ),
    Test = c("", "1 vs 2", "2 vs 3"),
    df = lrt_no_gender$df,
    logLik = lrt_no_gender$logLik,
    `L.Ratio` = lrt_no_gender$`L.Ratio`,
    `p-value` = lrt_no_gender$`p-value`,
    AIC = c(AIC(m_mean1), AIC(m_mean3), AIC(m_mean5)),
    BIC = c(BIC(m_mean1), BIC(m_mean3), BIC(m_mean5))
  ),
  tibble(
    Model = c(
      "4. Main effects (with gender)",
      "5. Two-way interactions (with gender)",
      "6. Three-way interactions (with gender)"
    ),
    Test = c("", "4 vs 5", "5 vs 6"),
    df = lrt_with_gender$df,
    logLik = lrt_with_gender$logLik,
    `L.Ratio` = lrt_with_gender$`L.Ratio`,
    `p-value` = lrt_with_gender$`p-value`,
    AIC = c(AIC(m_mean2), AIC(m_mean4), AIC(m_mean6)),
    BIC = c(BIC(m_mean2), BIC(m_mean4), BIC(m_mean6))
  )
) %>%
  mutate(
    across(c(logLik, `L.Ratio`, `p-value`, AIC, BIC), ~ round(.x, 3))
  )

paper_kable(
  mean_model_compare_all,
  "Likelihood ratio tests and information criteria for fixed-effects structure selection."
)
```

Table 11: Likelihood ratio tests and information criteria for fixed-effects structure selection.

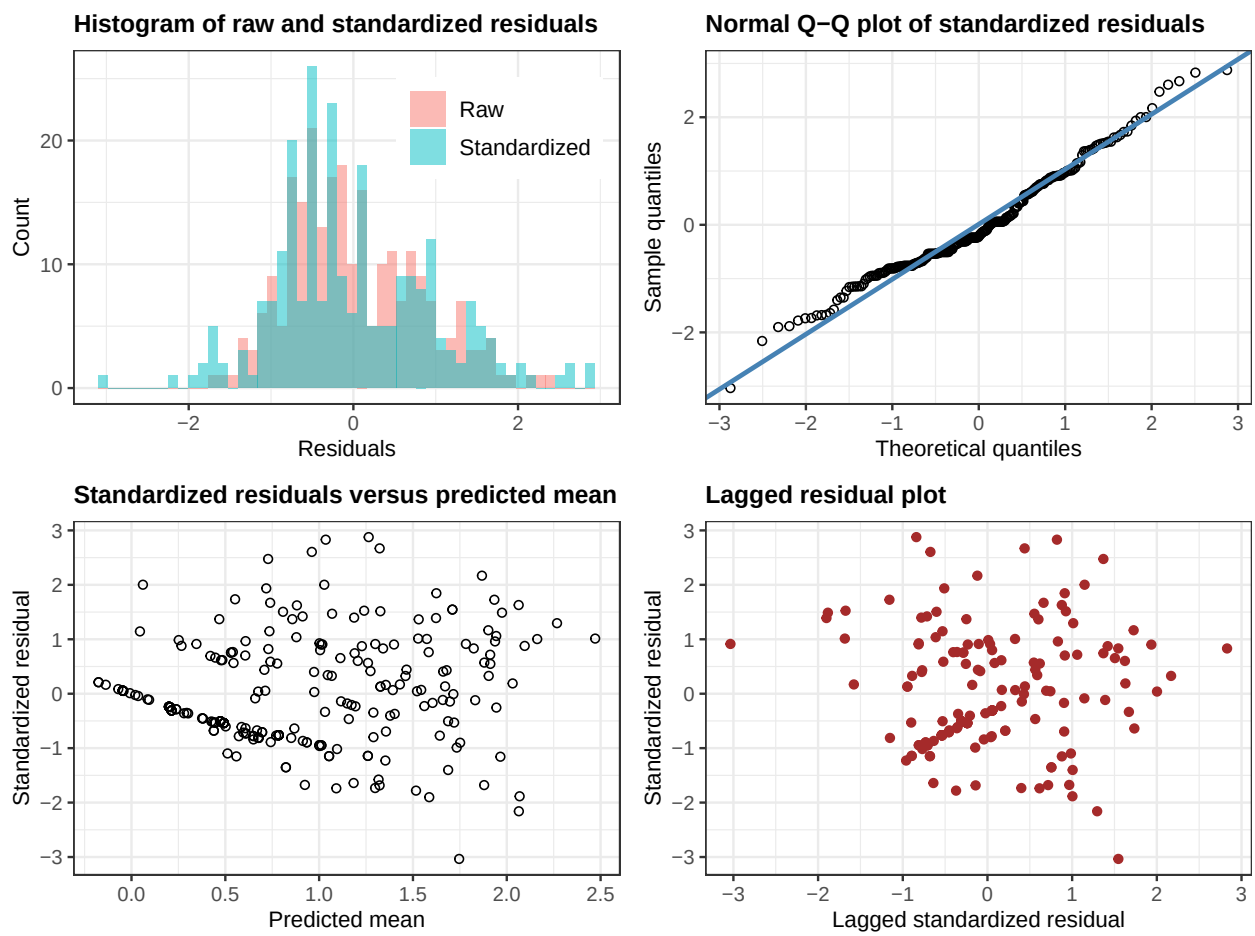
Model	Test	df	logLik	L.Ratio	p-value	AIC	BIC
1. Main effects (no gender)		7	-300.888	NA	NA	615.777	640.314
2. Two-way interactions (no gender)	1 vs 2	10	-298.333	5.110	0.164	616.667	651.720
3. Three-way interactions (no gender)	2 vs 3	11	-296.021	4.624	0.032	614.043	652.602
4. Main effects (with gender)		8	-299.052	NA	NA	614.103	642.146
5. Two-way interactions (with gender)	4 vs 5	14	-292.472	13.160	0.041	612.943	662.018
6. Three-way interactions (with gender)	5 vs 6	19	-288.797	7.349	0.196	615.594	682.196

Three way interaction without gender is the best model.

Model Diagnostic

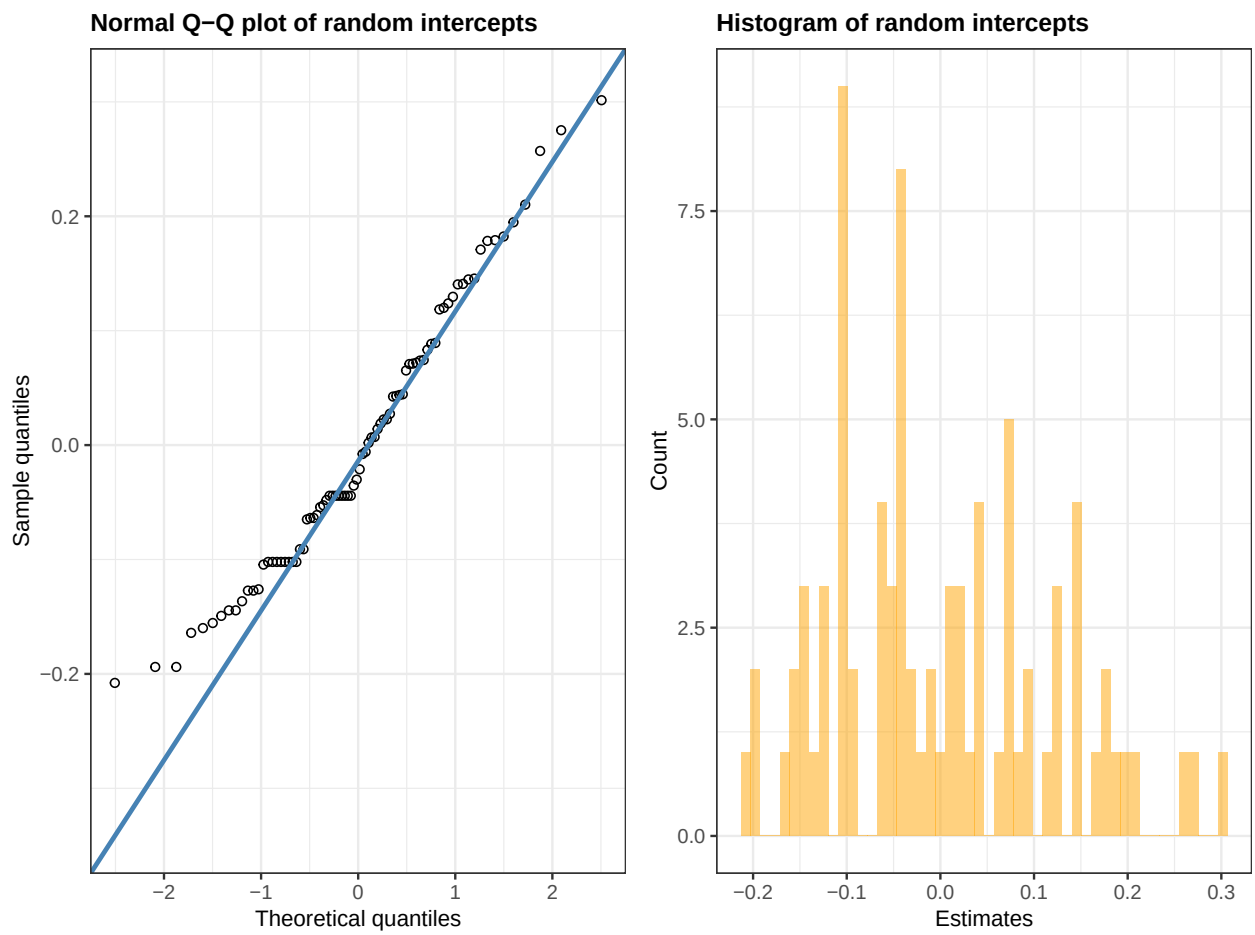
```
final_model_ar1 <- lme( # three way interaction model
  alcuse ~ age * coa * peer,
  random = ~ 1 | id,
  correlation = corAR1(form = ~ age | id),
  data = alc,
  method = "ML"
)

rhat <- resid(final_model_ar1, type = "normalized")

condmean <- predict(final_model_ar1)
y <- alc$alcuse
stg1r <- y - condmean

rtype <- c(rep("Raw", length(stg1r)), rep("Standardized", length(stg1r)))
resid <- c(stg1r, rhat)

stg1resids <- data.frame(
  rtype = rtype,
  resid = as.numeric(resid)
)

p1 <- ggplot(stg1resids, aes(x = resid, fill = rtype)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50) +
  labs(
    title = "Histogram of raw and standardized residuals",
    y = "Count",
    x = "Residuals",
    fill = NULL
  ) +
  theme(
    legend.position = c(0.78, 0.78)
  ) +
  base_plot_theme

p2 <- ggplot(data.frame(rhat = rhat), aes(sample = rhat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of standardized residuals",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

yhat <- predict(final_model_ar1)

p3 <- ggplot(data.frame(yhat = yhat, rhat = rhat), aes(y = rhat, x = yhat)) +
  geom_point(shape = 1) +
  labs(
    title = "Standardized residuals versus predicted mean",
```

```
    x = "Predicted mean",
    y = "Standardized residual"
  ) +
  base_plot_theme

p4 <- alc %>%
  mutate(rhat = rhat) %>%
  group_by(id) %>%
  mutate(rhat1 = lag(rhat)) %>%
  ggplot(aes(y = rhat, x = rhat1)) +
  geom_point(color = "brown", na.rm = TRUE) +
  labs(
    title = "Lagged residual plot",
    y = "Standardized residual",
    x = "Lagged standardized residual"
  ) +
  base_plot_theme

(p1 + p2) / (p3 + p4)
```




```
re <- ranef(final_model_ar1) %>%
  rownames_to_column("id")
```

```r
b0hat <- re$`(Intercept)`

p1 <- ggplot(data.frame(b0hat = b0hat), aes(sample = b0hat)) +
  stat_qq(shape = 1) +
  stat_qq_line(color = "steelblue", linewidth = 1) +
  labs(
    title = "Normal Q-Q plot of random intercepts",
    x = "Theoretical quantiles",
    y = "Sample quantiles"
  ) +
  base_plot_theme

p3 <- ggplot(data.frame(b0hat = b0hat), aes(x = b0hat)) +
  geom_histogram(position = "identity", alpha = 0.5, bins = 50, fill = "orange") +
  labs(
    title = "Histogram of random intercepts",
    x = "Estimates",
    y = "Count"
  ) +
  base_plot_theme


(p1 + p3)
```



Model interpretation

```r
# print the result of the chosen model: three way interaction model
coef_table <- summary(final_model_ar1)$tTable %>%
  as.data.frame() %>%
  rownames_to_column("Variable") %>%
  rename(
    Estimate = Value,
    SE = Std.Error,
    DF = DF,
    `p-value` = `p-value`
  ) %>%
  mutate(
    `95% CI` = paste0(
      "(",
      round(Estimate - 1.96 * SE, 3),
      ", ",
      round(Estimate + 1.96 * SE, 3),
      ")"
    )
  ) %>%
  mutate(
    across(c(Estimate, SE, DF, `p-value`), ~ round(.x, 3))
  ) %>%
  mutate(
    Variable = dplyr::recode(
      Variable,
      "(Intercept)" = "Constant",
      "age" = "Age",
      "coa1" = "Parental alcoholism",
      "peer" = "Peer alcohol use",
      "age:coa1" = "Age × parental alcoholism",
      "age:peer" = "Age × peer alcohol use",
      "coa1:peer" = "Parental alcoholism × peer alcohol use",
      "age:coa1:peer" = "Age × parental alcoholism × peer alcohol use"
    )
  ) %>%
  select(Variable, Estimate, SE, `95% CI`, `p-value`)

paper_kable(
  coef_table,
  "Fixed effects estimates for the final mixed-effects model"
)
```

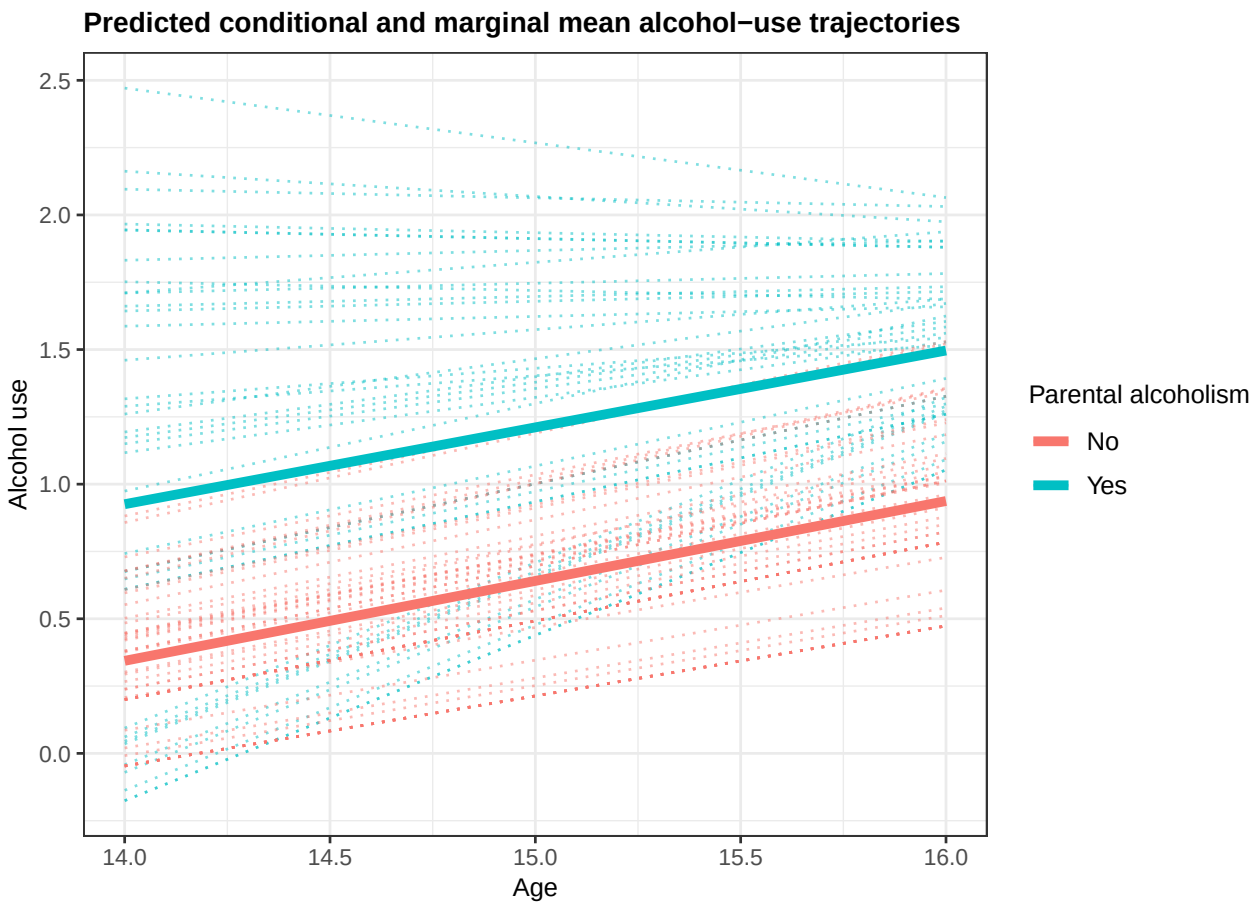
Prediction plot

```r
# compute conditional predicted values from the final model
condmean <- predict(final_model_ar1, level = 1)

# compute the mean peer alcohol use
```

Table 12: Fixed effects estimates for the final mixed-effects model

Variable	Estimate	SE	95% CI	p-value
Constant	-3.645	2.056	(-7.674, 0.385)	0.078
Age	0.260	0.137	(-0.007, 0.528)	0.059
Parental alcoholism	-5.041	3.176	(-11.267, 1.185)	0.117
Peer alcohol use	-0.164	1.817	(-3.725, 3.398)	0.928
Age × parental alcoholism	0.354	0.211	(-0.059, 0.768)	0.095
Age × peer alcohol use	0.036	0.121	(-0.201, 0.273)	0.766
Parental alcoholism × peer alcohol use	5.682	2.527	(0.73, 10.634)	0.027
Age × parental alcoholism × peer alcohol use	-0.359	0.168	(-0.688, -0.03)	0.034

```r
peer0 <- mean(alc$peer, na.rm = TRUE)

# create new data for marginal predictions across age and coa
margdat <- expand.grid(
  age = sort(unique(alc$age)),
  coa = levels(alc$coa),
  peer = peer0
)

# compute marginal predicted mean alcohol use
margdat$margmean <- predict(final_model_ar1, newdata = margdat, level = 0)

# combine predicted values with the original dataset
predalc <- as.data.frame(cbind(alc, condmean))

# plot of predicted alcohol use trajectories
ggplot(
  predalc,
  aes(x = age, y = condmean, group = id)
) +
  geom_line(aes(color = coa), linetype = 3, alpha = 0.5) +
  geom_line(
    data = margdat,
    aes(x = age, y = margmean, color = coa, group = coa),
    linewidth = 1.8,
    inherit.aes = FALSE
  ) +
  scale_color_manual(values = coa_cols, labels = c("No", "Yes"),
                     name = "Parental alcoholism") +
  labs(
    title = "Predicted conditional and marginal mean alcohol-use trajectories",
    x = "Age",
    y = "Alcohol use"
  ) +
  base_plot_theme
```


Predicted conditional and marginal mean alcohol−use trajectories
Alcohol use
2.5
2.0
1.5
1.0
0.5
0.0
14.0
14.5
15.0
15.5
16.0
Age
Parental alcoholism
No
Yes